

SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

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Agent Skills Remain Static After Deployment — Users Rediscover Solutions

Static Skill Paradigm

- Users manually install skills from a centralized hub
- Solutions discovered during interaction rarely persist beyond sessions
- Same failure and recovery patterns repeat across users
- Memory-based methods stay tied to specific instances
- Skill libraries treated as static resources

Needed: Evolving Skills

- Runtime experience should continuously improve skills
- Cross-user knowledge must accumulate at the system level
- Generalizable improvements separated from idiosyncratic fixes
- Skills adapt to diverse contexts without manual curation
- Closed-loop feedback drives autonomous improvement

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



1 Evidence

Analyze trajectories to identify behavioral patterns

2 Attribution

Diagnose root causes of failures and successes

3 Evolution

Propose skill mutations via open-ended reasoning

4 Distribution

Verify, merge, and synchronize to all agents

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

01

Evidence

Analyze structured session trajectories to identify behavioral patterns across action–feedback causal chains.



02

Attribution

Diagnose root causes of failures and successes by reasoning over aggregated cross-user interaction evidence.



03

Evolution

Propose skill mutations—refining existing skills or creating new ones—through open-ended agentic reasoning, not predefined rules.

Key insight: The evolver performs autonomous open-ended reasoning over interaction evidence—not predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why Cross-User Aggregation Matters

- Different users exercising the same skill under diverse contexts produce complementary views of its behavioral boundary
- Reveals both conditions where the skill works and where it breaks
- A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix
- Structured session trajectories preserve full action–feedback causal chains for aggregation

Compatible Claw-Style Systems

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

Evaluation on WildClawBench using Qwen3-Max demonstrates substantial improvements across tasks.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Day	Candidate Skill	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Reject	Not admitted to next cycle

Result: Only 1 of 6 candidates accepted — illustrating the conservative verification process that prevents regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no	Reject	Not admitted to next cycle

Result: 3 of 6 days yielded accepted updates — showing successful iterative refinement before reaching a plateau.

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing

01

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Cross-user trajectory aggregation provides complementary behavioral signals that a single user cannot generate alone.

02

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system improves while users work, not through post-hoc review.

03

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over interaction evidence rather than predefined update rules. The evolver autonomously diagnoses and mutates skills.

Implication: Continuously improving agent systems are now feasible through evidence-driven, autonomous skill evolution at scale.